

Dept: Computer Science Engineering
Subject: Discrete Math.

Question : 2019-2020 (Section-A)

Questions Answers of 2019

Answer to the Q. no: 1(a)

proposition: A proposition is a collection of declarative statements that has either a truth value 'true' or a truth value 'false'.

Example: $5+2=7$, He is a boy.

Predicate Calculus: The branch of symbolic logic that uses symbols for quantifiers and for arguments and predicates of propositions as well as for ~~an~~ unanalyzed propositions and logical connectives, is called Predicate calculus.

Example:

~~$\forall x \in P(x)$~~ $P(x) = x$ can speak English
 $Q(x) = x$ can speak Bangla

Statement: There is a student at your school who can speak English and who knows also Bangla.

$\therefore$ conclusion: $\exists x (P(x) \wedge Q(x))$

Answer to the Q. No: 1 (v)

Given, If it is raining then I get wet

converse: If I get wet then it is raining.

Inverse: If it is not raining then I don't get wet.

contrapositive: If I don't get wet then it is not raining.

Answer to the Q. No: 1 (c)

P	Q	R	$\neg P$	$(P \rightarrow Q)$	$(P \rightarrow Q) \rightarrow R$	$((P \rightarrow Q) \rightarrow R) \vee \neg P$
T	T	T	F	T	T	T
T	T	F	F	T	F	F
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	F	T
F	F	T	T	T	T	T
F	F	F	T	T	F	T

Thus, every truth value of the expression is not true.
Therefore $\neg \vdash$

Answer to the Q. NO: 2 (a)

Given, $P \wedge (P \rightarrow Q)$

Disjunctive Normal Form, $P \wedge (P \rightarrow Q)$

$$= P \wedge (\neg P \vee Q)$$

$$= (P \wedge \neg P) \vee (P \wedge Q)$$

or symbol ante hobe

And,

Conjunctive Normal form, $P \wedge (P \rightarrow Q)$

and symbol ante hobe

$$= P \wedge (\neg P \vee Q)$$

Answer to the Q. NO: 2 (b)

Given, $\forall x \exists y [P(x, y) \wedge Q(x, y) \rightarrow R(x, y)]$

By Negating we get,

$$\neg (\forall x \exists y [P(x, y) \wedge Q(x, y) \rightarrow R(x, y)])$$

$$= \exists x \forall y \neg [\neg (P(x, y) \wedge Q(x, y)) \vee R(x, y)]$$

$$= \exists x \forall y [(P(x, y) \wedge Q(x, y)) \wedge \neg R(x, y)]$$

which is the negation of given statement.

Answer to the Q. NO: 3(a)

Assume,

$$p = 3n+2 \text{ is odd}$$

$$q = n \text{ is even.}$$

$\therefore 3n+2$ is odd and n is even.

$$\text{So, } n = 2k$$

$$\text{Now, } 3n+2 = 3(2k)+2$$

$$= 6k+2$$

$$= 2(3k+1)$$

$$= 2t$$

$$\boxed{3k+1 = t}$$

$$\therefore 3n+2 = \text{even}$$

This is contradiction.

Hence, n should be odd.

Answering Questions of 2020

Answer to the Q. No: 1(a)

A proposition is a collecting of declarative statements that ~~has~~ either a truth value "truth" or a truth value "false".

Example: "man is mortal", (it returns truth value)

① "Colombo is the capital of Pakistan".

It returns false. so, it's a proposition.

② " $2+3=5$ "

It returns true. so, it's a proposition.

③ " $x+2=11$ "

It ~~returns~~ not returns any truth value, so, it's not a proposition.

④ " $x+y=y+x$ "

It returns ~~truth~~ true. so, it is a proposition.

Answer to the Q.NO: 1(b)

Given Statement,

"The home team wins whenever it is raining?"

Contrapositive: If the home team doesn't win, then it is not raining.

Converse: If the home team wins, then it is raining.

Inverse: If it is not raining then the home team doesn't win.

Answer to the Q.NO: 1(c)

P	Q	$(P \rightarrow Q)$	$\neg P$	$(\neg P \vee Q)$	$(P \rightarrow \neg P) \leftrightarrow (\neg P \vee Q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

Thus, here a truth value of the statement is true for every value's of its propositional variable. So, it's a tautology.

Answer to the Q. no: 1(d)

Given statement,

"You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."

let,

p = you can ride the roller coaster.

q = you are under 4 feet tall.

r = you are older than 16 years old.

logical Expression: $(q \wedge \neg r) \rightarrow \neg p$

~~$(\neg p \rightarrow q) \rightarrow \neg r \rightarrow \neg p$~~

Answer to the Q. no: 2(a)

Given Statement:

"Every student in this class has studied calculus"

let, $Q(x) = x$ studied calculus.

$P(x) = x$ student of the class.

$\forall x$ = for every student.

$\therefore$ ~~$\forall x$~~ predicates and quantifiers expression:

~~$\forall x (P(x))$~~ $\forall x (P(x) \rightarrow Q(x))$

Answer to the Q. No: 2(b)

Given,

$$Q(x, y) \Rightarrow x+y=0$$

His course all set of real numbers.

$$\therefore \forall x \exists y Q(x, y) = \forall x \exists y Q(x+y=0)$$

which means,

"for every real number x , there exists a y such that $x+y=0$ ". which is True,

Because for specific x there is y which can make $x+y=0$.

And,

$$\exists y \forall x Q(x, y) \equiv \exists y \forall x Q(x+y=0)$$

which means,

"there is some real number y for every real number of x . which is not true.

Because for specific ~~numb~~ real number y there is not ~~no~~ exists x for every real number. so, it's can't make $x+y=0$.

Answer to the Q. NO: 2(c)

Given,

$M(x, y) = x$ has sent y an e-mail message

$T(x, y) = x$ has ~~has~~ telephoned y

Expressions:

① $\forall x \exists y M(x, y)$

① $\forall x M(x, \text{kamal})$

② $\forall x \exists y \neg T(x, y)$

② $\forall x (\neg T(x, \text{Arup}) \vee M(x, \text{Arup}))$

③ $\forall x \exists y T(x, y) \vee M(x, y)$

③ $\forall x \neg T(x, \text{Nandini})$

④ $\exists y \forall x M(x, y) \vee T(x, y)$

④ $\exists x \forall y (y \neq x \rightarrow (M(x, y) \vee T(x, y)))$

Answer to the Q. NO: 3(a)

Rules of Inference: Rules of inference are group of logical expressions form or guide consisting of premises and draws a conclusion.

Here, $M(x) = x$ is mortal.

$N(x) = x$ is a man.

And,

premise-1: "All men are mortal"

$\therefore \forall x (N(x) \rightarrow M(x))$

premise-2: "Socrates is ~~mortal~~ a man"

Conclusion: "Socrates is mortal"

$\therefore M(\text{Socrates})$

Now,

Step (1): $\forall x (N(x) \rightarrow M(x))$ [Premise-1]

Step (2): $N(\text{Socrates}) \rightarrow M(\text{Socrates})$ [Universal instantiation form (1)]

Step (3): $N(\text{Socrates})$ [Premise-2]

$\therefore M(\text{Socrates})$ [2, 3, Modus Ponens]

Answer to the Q. No: 3(b)

let,

P = You send me an email message.

Q = I will finish writing the program.

R = I will go to sleep.

S = I will wake up feeling refreshed.

Premise 1: $P \rightarrow Q$

Premise 2: $\neg P \rightarrow R$

Premise 3: $R \rightarrow S$

Conclusion: $\neg Q \rightarrow S$

$$(ii) \forall y P(1, y)$$

$\forall$ - means all

Here, $y = 1, 2, 3$

Now,

$$\forall y P(1, y) = P(1, 1) \wedge P(1, 2) \wedge P(1, 3)$$

Therefore, proposition using disjunction and conjunction

$$\text{is } P(1, 1) \wedge P(1, 2) \wedge P(1, 3)$$

$$(iii) \exists y \neg P(2, y)$$

$\exists$ - means there exists

$\neg$ - means negation

Here, $y = 1, 2, 3$

$$\text{Now, } \exists y \neg P(2, y) = \neg P(2, 1) \vee \neg P(2, 2) \vee \neg P(2, 3)$$

Therefore, proposition using disjunction and conjunction

$$\text{is } \neg P(2, 1) \vee \neg P(2, 2) \vee \neg P(2, 3)$$

$$(iv) \forall x \neg P(x, 2)$$

$\forall x$ - means all

$\neg$ - negation

Here, $x = 1, 2, 3$

$$\text{Now, } \forall x \neg P(x, 2) = \neg P(1, 2) \wedge \neg P(2, 2) \wedge \neg P(3, 2)$$

Therefore, proposition using disjunction and conjunction

$$\text{is } \neg P(1, 2) \wedge \neg P(2, 2) \wedge \neg P(3, 2)$$

- Step 1: ~~$\neg P \rightarrow \neg P$~~ $\neg P \rightarrow r$ [Premis - 1]
- Step 2: $r \rightarrow s$ [Premise - 2]
- Step 3: $\neg P \rightarrow s$ [1, 2 Hypothetical syllogism]
- Step 4: $p \rightarrow q$ [Premise 1]
- Step 5: $q \rightarrow p$ [5, converse]
- Step 6: ~~$\neg p$~~ $p \rightarrow \neg s$ [3, inverse]
- Step 7: $q \rightarrow \neg s$ [5, 6 Hypothetical syllogism]
- Step 8: $\neg q \rightarrow s$ [7, inverse]

Answer to the Q. No: 3 (c)

We have the domain of the proposition function $P(x, y)$ consists of pair x and y , where x is 1, 2, 3 and y is 1, 2, 3.

Now,

$$(i) \exists x P(x, 3)$$

$\exists$ - means there exists

Here, $x = 1, 2, 3$

Now,

$$\exists x P(x, 3) = P(1, 3) \vee P(2, 3) \vee P(3, 3)$$

Therefore, per proposition using disjunction and conjunction is $P(1, 3) \vee P(2, 3) \vee P(3, 3)$